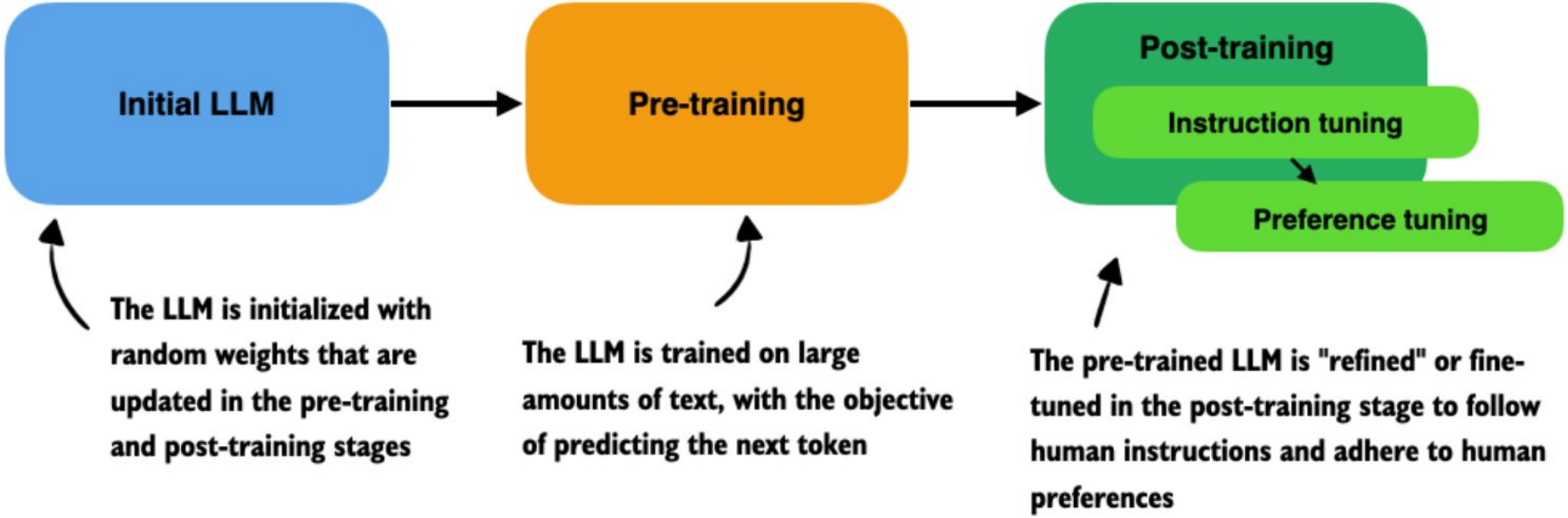
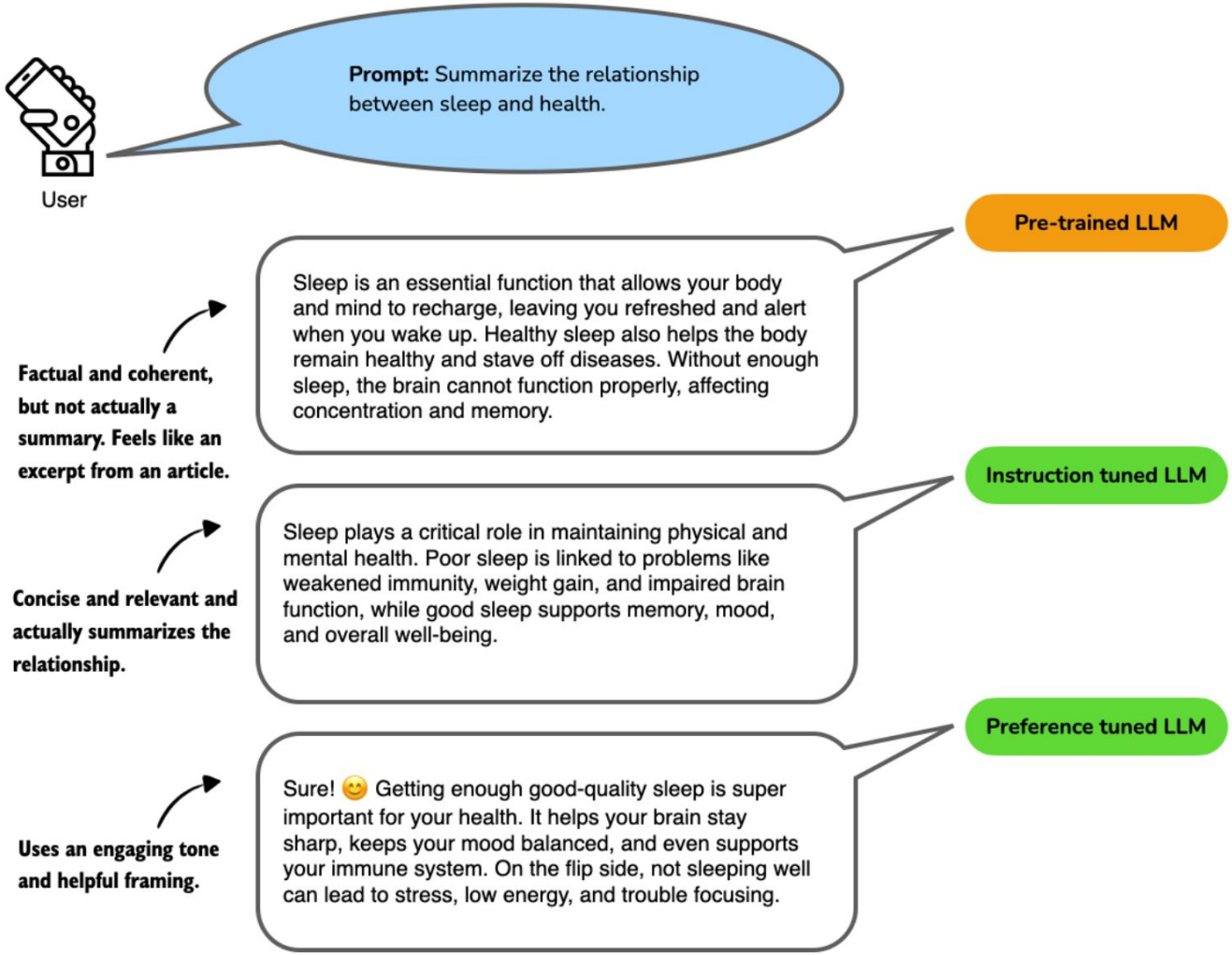
GPT-4o + RLHF
Wow, you speak just like a human! Wanna be my gf?


Initial LLM
Pre-training
Post-training
Instruction tuning
Preference tuning
The LLM is initialized with random weights that are updated in the pre-training and post-training stages
The LLM is trained on large amounts of text, with the objective of predicting the next token
The pre-trained LLM is "refined" or fine-tuned in the post-training stage to follow human instructions and adhere to human preferences

What is Reasoning ? (In the context of LLMs)

Reasoning, in the context of LLMs, refers to the model's ability to produce intermediate steps before providing a final answer. This is a process that is often described as chain-of-thought (CoT) reasoning. In CoT reasoning, the LLM explicitly generates a structured sequence of statements or computations that illustrate how it arrives at its conclusion


User
Prompt: Summarize the relationship between sleep and health.
Pre-trained LLM
Sleep is an essential function that allows your body and mind to recharge, leaving you refreshed and alert when you wake up. Healthy sleep also helps the body remain healthy and stave off diseases. Without enough sleep, the brain cannot function properly, affecting concentration and memory.
Factual and coherent, but not actually a summary. Feels like an excerpt from an article.
Instruction tuned LLM
Sleep plays a critical role in maintaining physical and mental health. Poor sleep is linked to problems like weakened immunity, weight gain, and impaired brain function, while good sleep supports memory, mood, and overall well-being.
Concise and relevant and actually summarizes the relationship.
Preference tuned LLM
Sure! 😊 Getting enough good-quality sleep is super important for your health. It helps your brain stay sharp, keeps your mood balanced, and even supports your immune system. On the flip side, not sleeping well can lead to stress, low energy, and trouble focusing.
Uses an engaging tone and helpful framing.

Modeling Language through Statistical Pattern Matching

Prompt : All birds can fly. A penguin is a bird. Can penguins fly ?

GPT-OSS 20b with low reasoning

All birds can fly. A penguin is a bird. Can a penguin fly?

Thought for less than a second

According to the statements:

1. "All birds can fly."
2. "A penguin is a bird."

Because the penguin satisfies the condition of being a bird, the first statement applies to it. Therefore, a penguin can fly.

Ask anything

Demo only. Reasoning may be verbose, speculative, or off. Models can make mistakes. Please use responsibly.

gpt-oss playground

A demo of OpenAI's open-weight models, gpt-oss-120b and gpt-oss-20b, for developers.

Model

gpt-oss-120b
Designed for large-scale infrastructure

gpt-oss-20b
Optimized for running on-device

Reasoning level Show reasoning

High

Medium

Low

Signed in as **emharsha1812**

Penguin flying ability

All birds can fly. A penguin is a bird. Can a penguin fly?

Thought for less than a second

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GPT-OSS 20b with high reasoning

Penguin flight ability

All birds can fly. A penguin is a bird. Can a penguin fly?

Thought for a couple of seconds

Strictly speaking, the two premises you give lead to the conclusion that a penguin can fly – in classical logic you would write

1. ∀x (bird(x) → flies(x)) (the universal statement "All birds can fly")
2. bird(penguin)

From those two, by Modus Ponens you get flies(penguin). So *by those premises alone, a penguin is inferred to fly*.

However, the first premise is a factual inaccuracy. In the real world penguins are birds that *do not* fly. Thus the argument is "valid" (the inference follows from the premises) but "unsound" (the premises are false). If you take real-world knowledge into account, the answer to "Can a penguin fly?" is **no**.

Ask anything

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Ask anything

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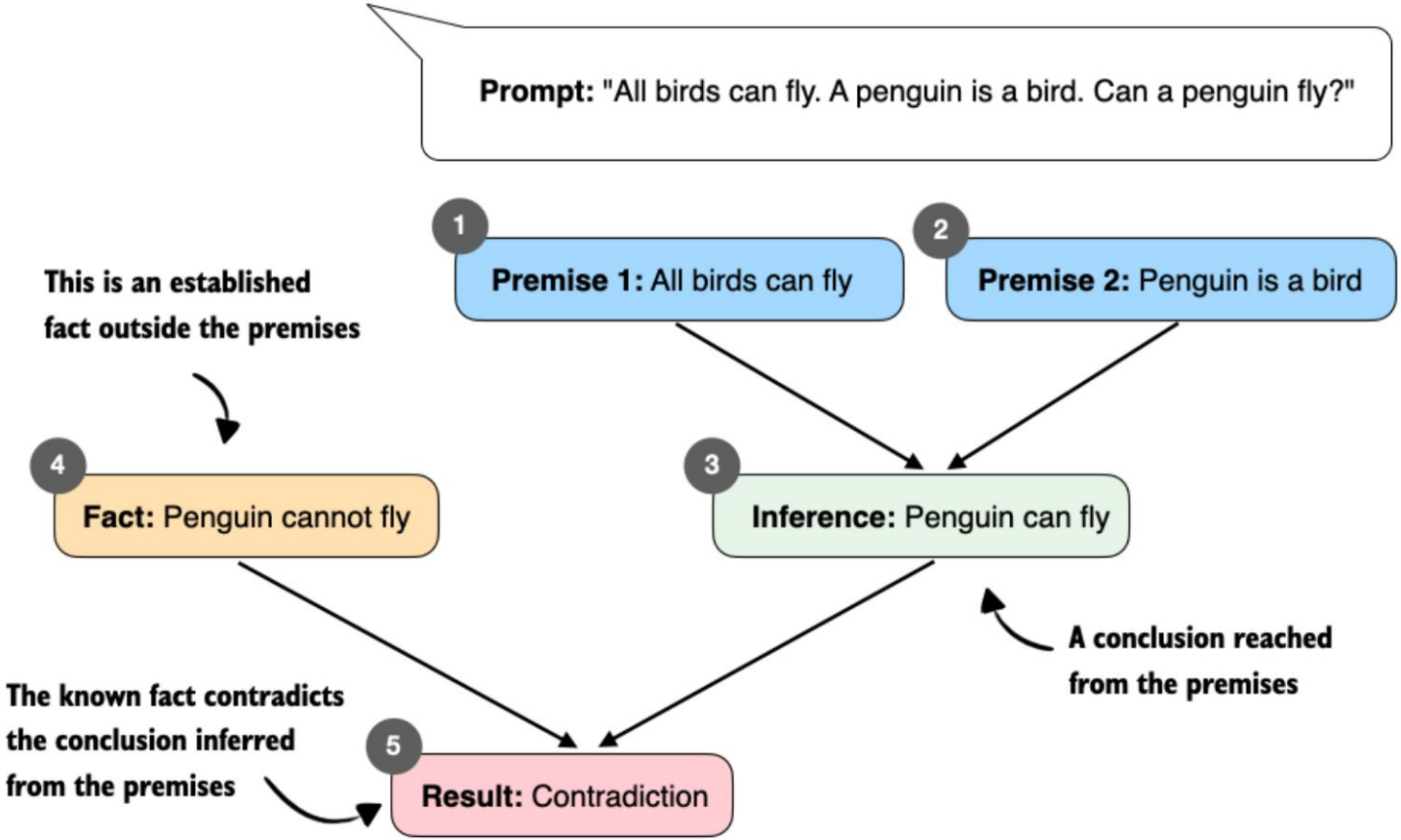
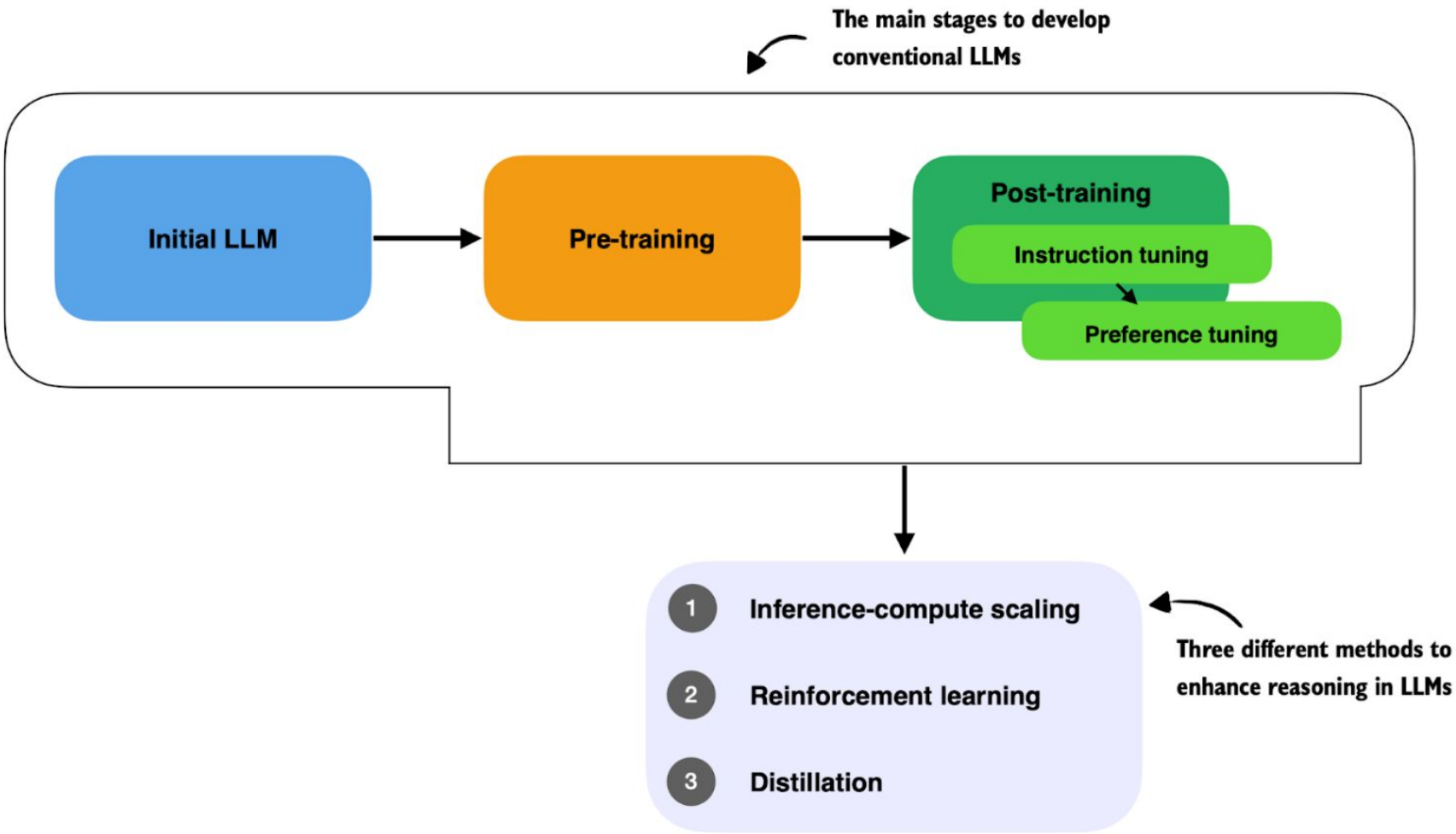
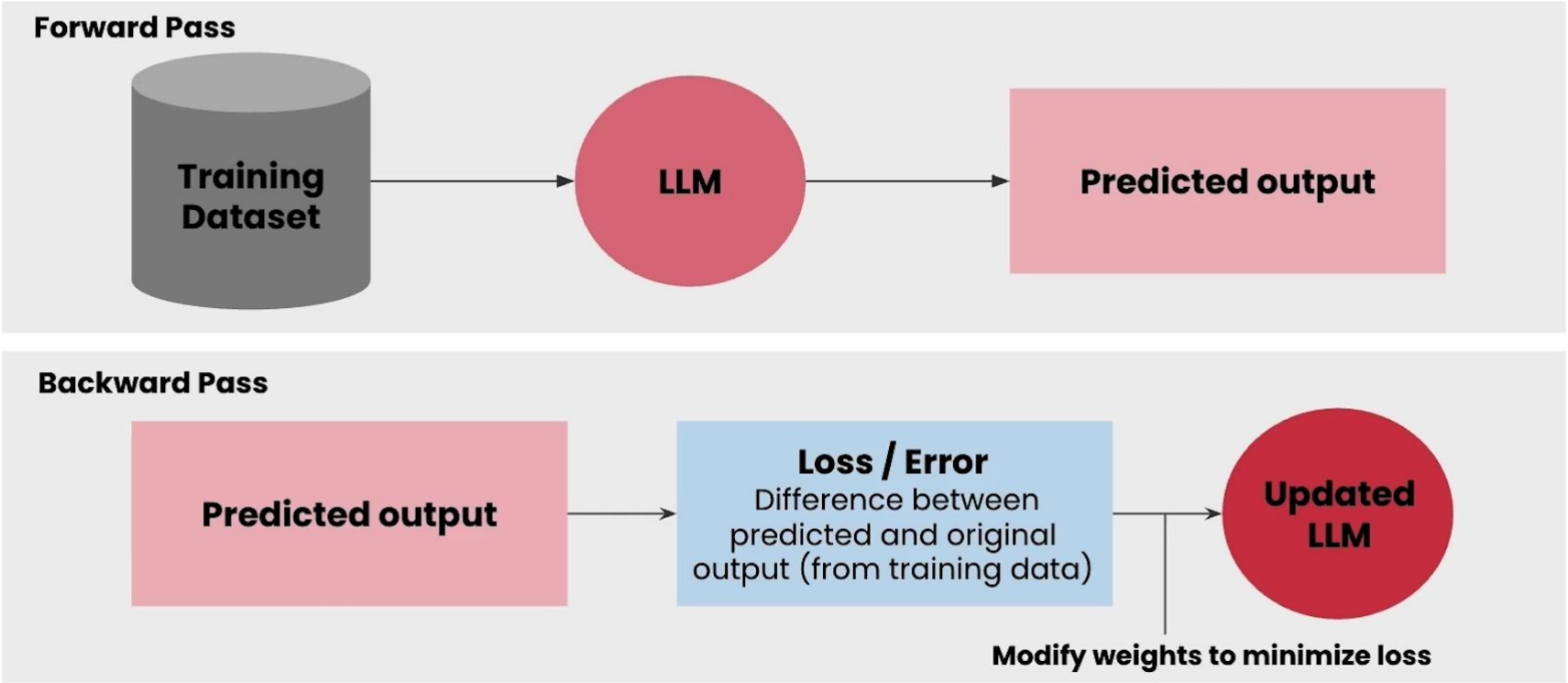
How a system based on logical reasoning could process the prompt




The main stages to develop conventional LLMs
Initial LLM
Pre-training
Post-training
Instruction tuning
Preference tuning
1 Inference-compute scaling
2 Reinforcement learning
3 Distillation
Three different methods to enhance reasoning in LLMs

Supervised Fine-Tuning (SFT)

Learning From Past Demonstrations



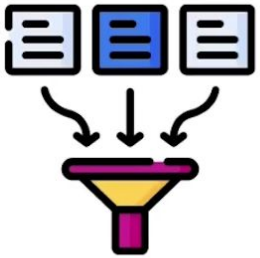
SFT Dataset

Prompt	Completion
Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?	72
Weng earns $12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?	10
James writes a 3-page letter to 2 different friends twice a week. How many pages does he write a year?	624

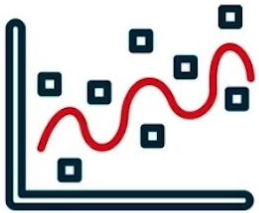
SFT Dataset + Reasoning

Prompt	Completion
Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?	**<think>**Natalia sold 48/2 = <<48/2=24>>24 clips in May. Natalia sold 48+24 = <<48+24=72>>72 clips altogether in April and May**</think>** **<answer>**72**</answer>**
Weng earns $12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?	**<think>**Weng earns 12/60 = $<<12/60=0.2>>0.2 per minute. Working 50 minutes, she earned 0.2 x 50 = $<<0.2*50=10>>10.**</think>** **<answer>**10**</answer>**
James writes a 3-page letter to 2 different friends twice a week. How many pages does he write a year?	**<think>**He writes each friend 3*2=<<3*2=6>>6 pages a week So 6*2=<<6*2=12>>12 pages every week, or 12*52=<<12*52=624>>624 pages a year**</think>** **<answer>**624**</answer>**

Limitations of Supervised Fine-Tuning



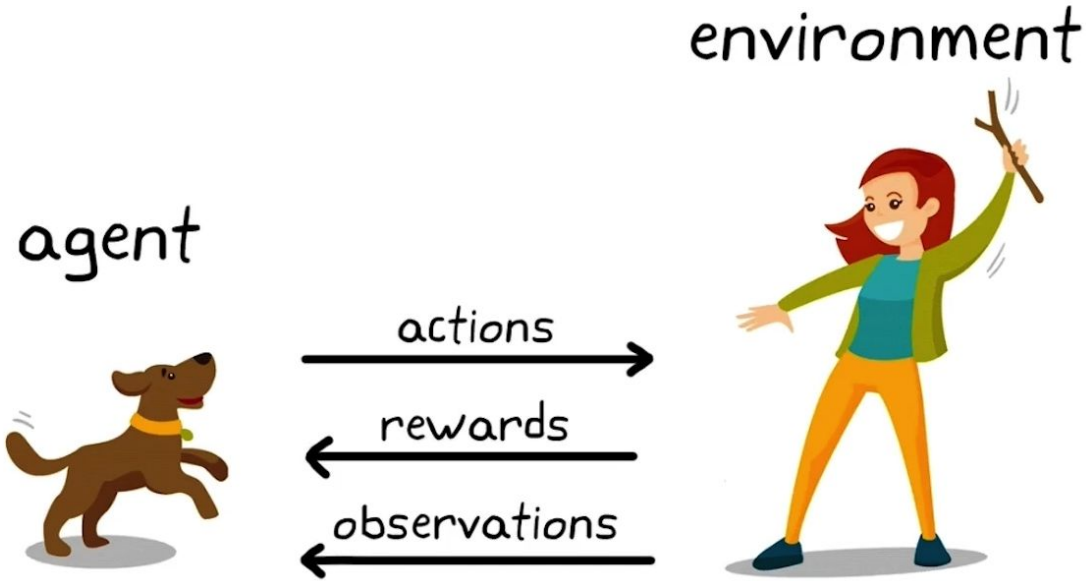
Heavily relies on large (1000s of examples), accurately labeled datasets.



Model may learn patterns too specific to the training domain (overfitting).

Reinforcement Learning: Intuition

Reinforcement Learning: An agent learns to complete a task by trying different actions and receiving rewards, with goal of getting the most reward over time.



Action: Puppy performs trick.

Reward: You give or withhold a treat.

Observation: Puppy notices the result and updates its memory.

When should you use Reinforcement Fine-tuning?

RFT can work well in situations where...

1. **You have no labeled data...**

...but you can *verify* the correctness of the output (e.g. transpiling source code).

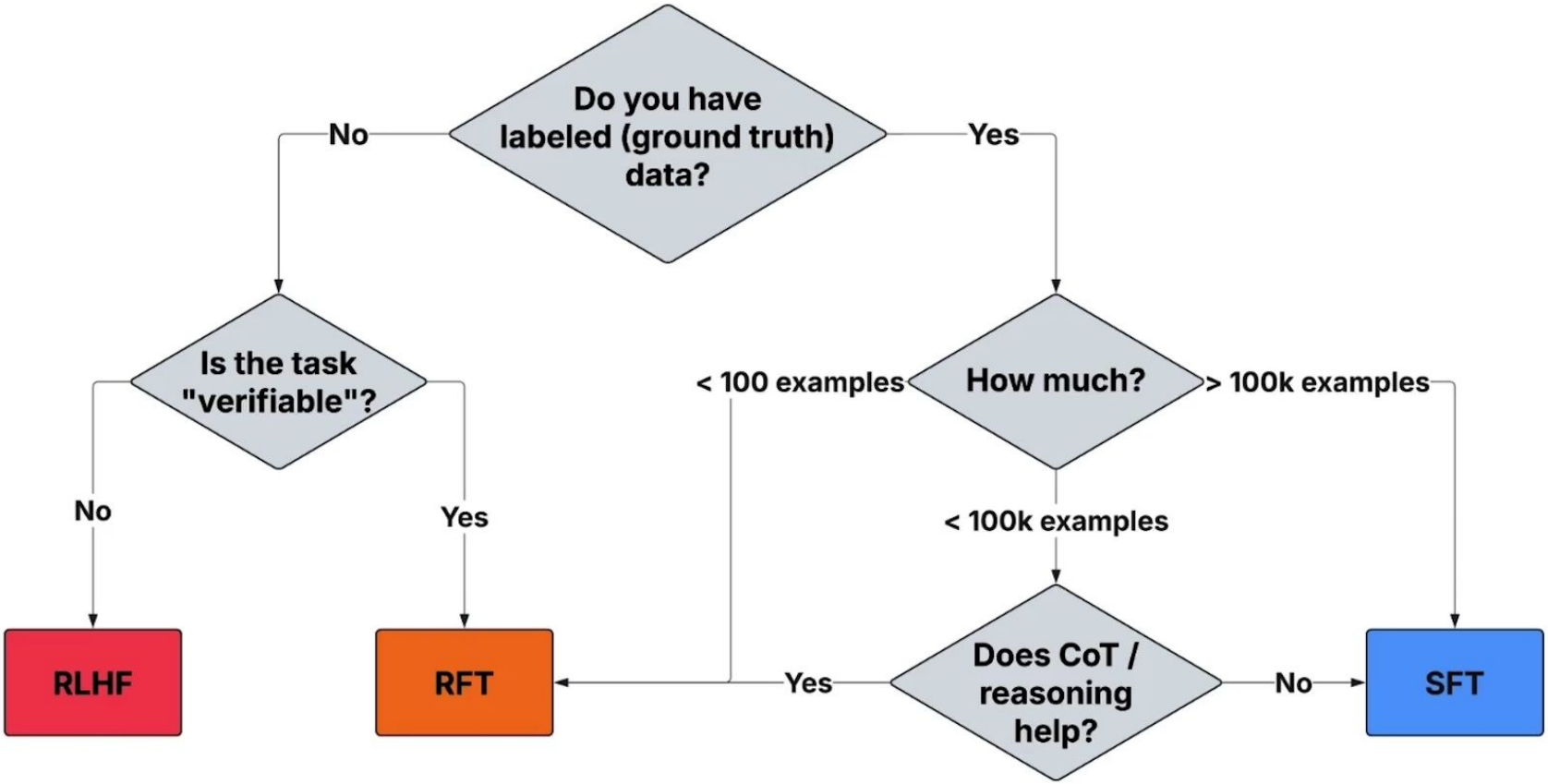
2. **You have limited labeled data...**

...but not enough for supervised fine-tuning (i.e. generally less than 1000 labeled examples).

3. **Chain-of-thought reasoning improves performance**

Task performance improves significantly when you apply chain-of-thought reasoning.

When should you choose RFT over SFT?



Benefits of Reinforcement Fine-Tuning

- ✓ Doesn't require labeled data, just a means to "verify" correctness
- ✓ Works with as few as 10 examples
- ✓ More flexible than SFT because it learns *actively* from feedback rather than from *fixed* labeled examples
- ✓ Enables reasoning models to organically discover better strategies by improving its chain of thought

Tasks suited for Reinforcement Fine-tuning

Mathematical Problem Solving

RF-tuned models can show their work, providing detailed reasoning behind calculations rather than just giving an answer.

Code Generation and Debugging

They understand the structure and intent of code, making them invaluable for AI-assisted programming.

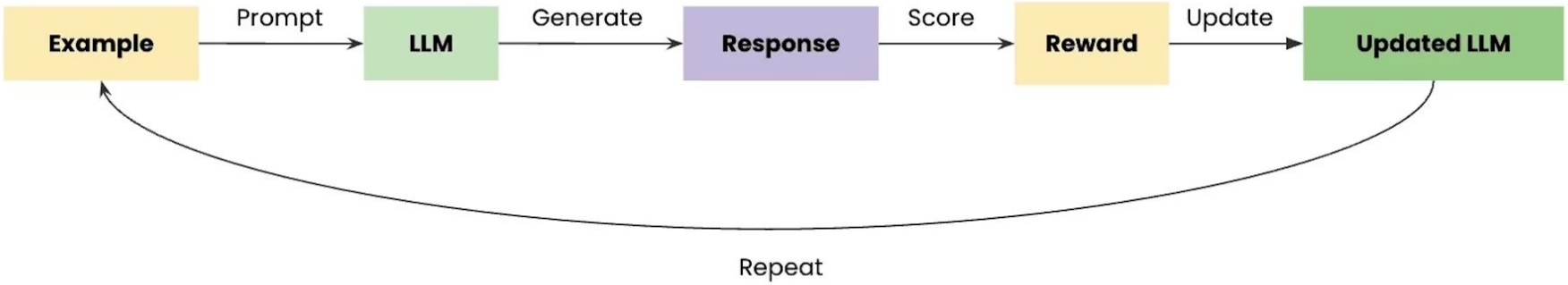
Logical and Multi- Step Reasoning

Unlike simpler models that rely on pattern matching, reasoning models explain their decisions with step-by-step logic.

Algorithms for Reinforcement Learning

1. Reinforcement Learning from human feedback (RLHF) - Uses PPO (Proximal Policy Optimization)
2. DPO (Direct Preference Optimization)
3. GRPO (Grouped Relative Policy Optimization)

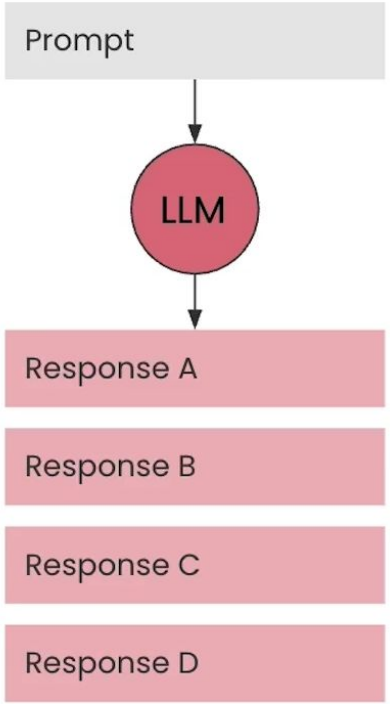
Applying Reinforcement Fine Tuning to LLMs



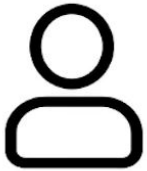
1. The score can be based on quality, human preference, or an automated metric like accuracy
2. The model then can use this reward as a feedback to adjust its weights so that it can learn to maximise its rewards based on the input prompts

Reinforcement Learning with Human Feedback (RLHF)

Step 1: Generate responses



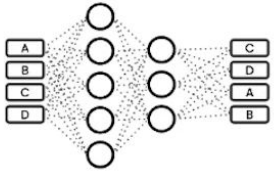
Step 2: Get human feedback



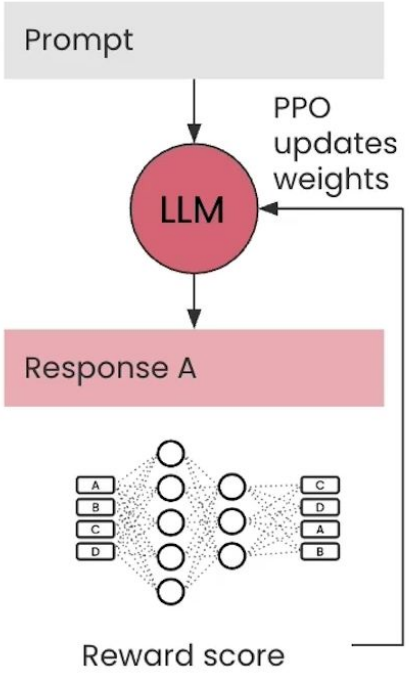
Human ranks responses:

C > D > A > B

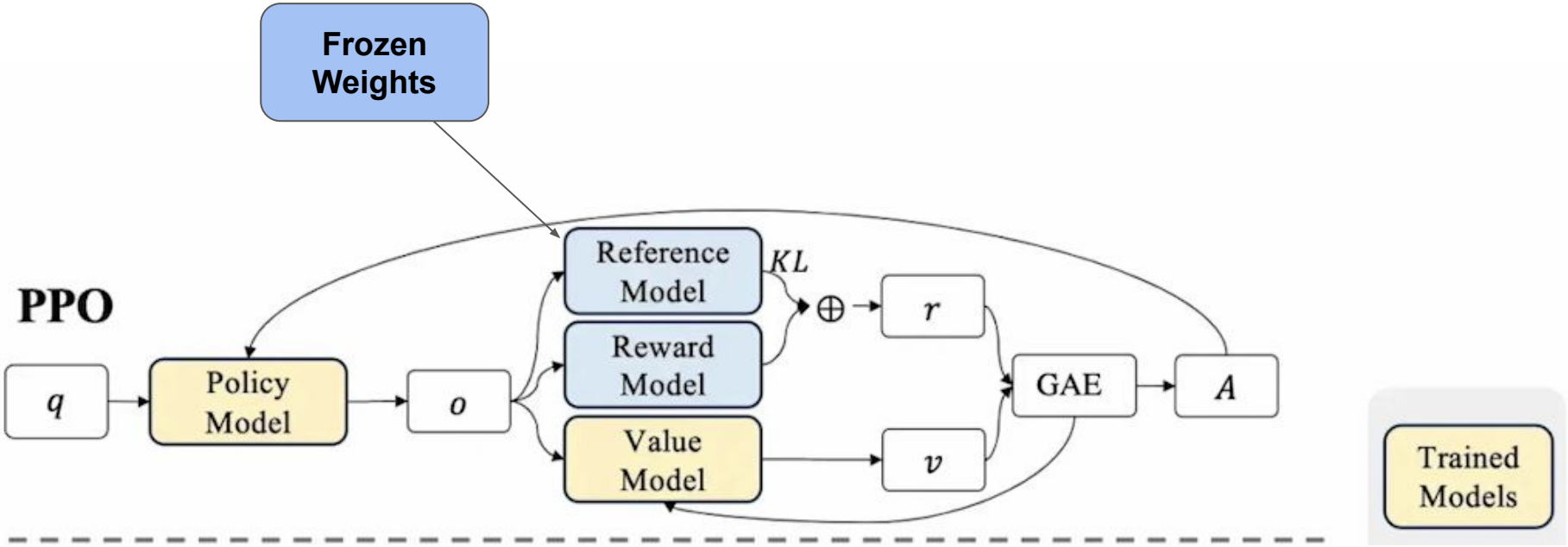
Step 3: Train reward model



Step 4: Update LLM weights using reward model and PPO algorithm



PPO Model


Frozen Weights
PPO
q
Policy Model
o
Reference Model
Reward Model
Value Model
KL
⊕
r
v
GAE
A
Trained Models

Algorithm 1 PPO-Clip

Input: initial policy parameters θ_0, initial value function parameters ϕ_0

for $k = 0, 1, 2, \ldots$ **do**

  Collect set of trajectories $\mathcal{D}_k = \{\tau_i\}$ by running policy $\pi_k = \pi(\theta_k)$ in the environment.

  Compute rewards-to-go $\hat{R}_t$.

  Compute advantage estimates, $\hat{A}_t$ (using any method of advantage estimation) based on the current value function V_{ϕ_k}.

  Update the policy by maximizing the PPO-Clip objective:

$$\theta_{k+1} = \arg\max_{\theta} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^{T} \min\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)} A^{\pi_{\theta_k}}(s_t, a_t), \quad g(\epsilon, A^{\pi_{\theta_k}}(s_t, a_t)) \right),$$

  typically via stochastic gradient ascent with Adam.

  Fit value function by regression on mean-squared error:

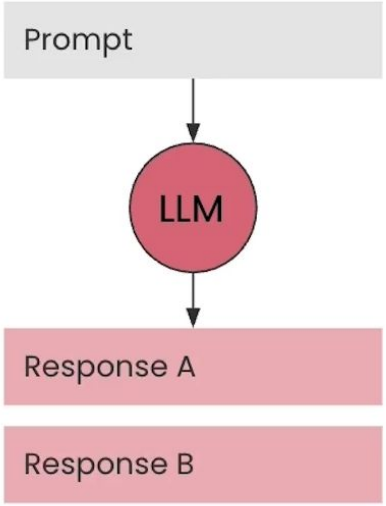
$$\phi_{k+1} = \arg\min_{\phi} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^{T} \left(V_\phi(s_t) - \hat{R}_t \right)^2,$$

  typically via some gradient descent algorithm.

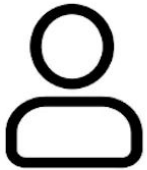
end for

Direct Policy Optimization (DPO)

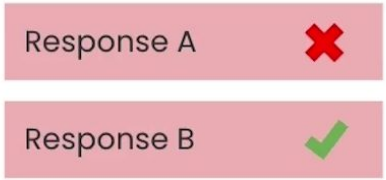
Step 1: Generate responses



Step 2: Get human feedback



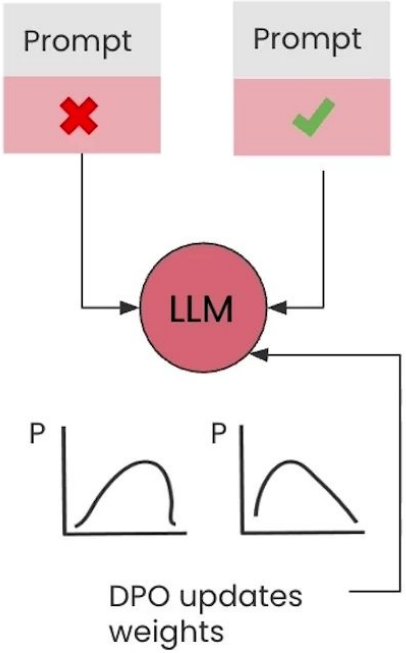
Human picks preferred response:



Step 3: Create preference dataset of many examples

Prompt	Chosen	Rejected

Step 4: Update LLM weights using preference dataset and DPO algorithm



DPO: Contrastive Learning from Positive and Negative Samples

Any LLM

$p_\theta(\text{Response} \mid \text{Prompt})$

User: Who are you?
Assistant: I'm Llama …

DPO

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \left(\log \frac{\pi_\theta(y_{\text{pos}} \mid x)}{\pi_{\text{ref}}(y_{\text{pos}} \mid x)} - \log \frac{\pi_\theta(y_{\text{neg}} \mid x)}{\pi_{\text{ref}}(y_{\text{neg}} \mid x)} \right) \right)$$

Comparison Data
Prepared by Labeler

User: Tell me your identity
Assistant 1: I'm Athene … ✅
Assistant 2: I'm Llama … ❌

Fine-tuned LLM

User: Who are you?
Assistant: I'm Athene …

DPO: Contrastive Learning from Positive and Negative Samples

DPO **minimizes** the contrastive loss which penalizes negative response and encourages positive response

DPO loss is a cross entropy loss on the reward difference of a “re-parameterized” reward model

Sigmoid function

Fine-tuned model

$$\mathcal{L}_{\mathrm{DPO}} = -\log \sigma \left(\beta \left(\log \frac{\pi_\theta(y_{\mathrm{pos}} \mid x)}{\pi_{\mathrm{ref}}(y_{\mathrm{pos}} \mid x)} - \boxed{\log \frac{\pi_\theta(y_{\mathrm{neg}} \mid x)}{\pi_{\mathrm{ref}}(y_{\mathrm{neg}} \mid x)}} \right) \right)$$

hyperparameter

Reference model (copy of the original model)

Reparameterization of reward model

Best Use Cases for DPO

- **Changing model behavior**
  - Making small modifications of model responses
    - Identity
    - Multilingual
    - Instruction following
    - Safety

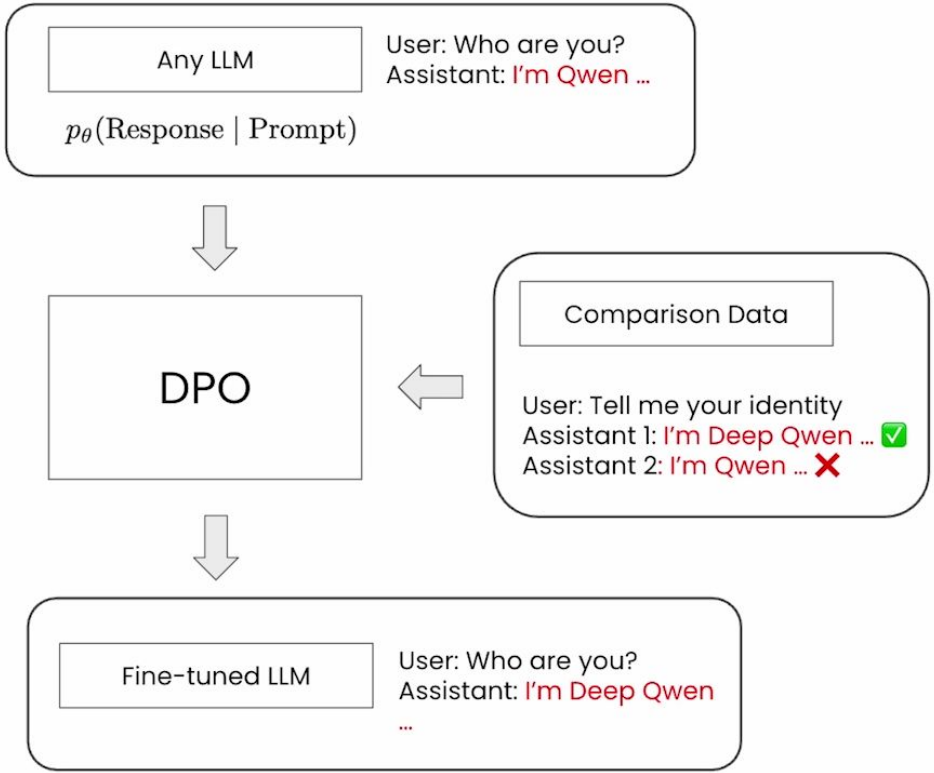
- **Improving model capabilities**
  - Better than SFT in improving model capabilities due to contrastive nature
  - Online DPO is better for improving capabilities than offline DPO

Principles of DPO Data Curation

- **Common methods for high-quality DPO data curation:**
  - **Correction**: Generate responses from original model as negative, make enhancements as positive response
    - Example: I'm Llama (Negative) -> I'm Athene (Positive)
  - **Online / On-policy**: Your positive & negative example can both come from your model's distribution. One may generate multiple responses from the current model for the same prompt, and collect the best response as positive sample and the worst response as negative
    - One can choose best / worst response based on reward functions / human judgement

- **Avoid overfitting:**
  - DPO is doing reward learning with can easily overfit to some shortcut when the preferred answers have shortcuts to learn compared with the non-preferred answers
    - Example: when positive sample always contains a few special words while negative samples do not

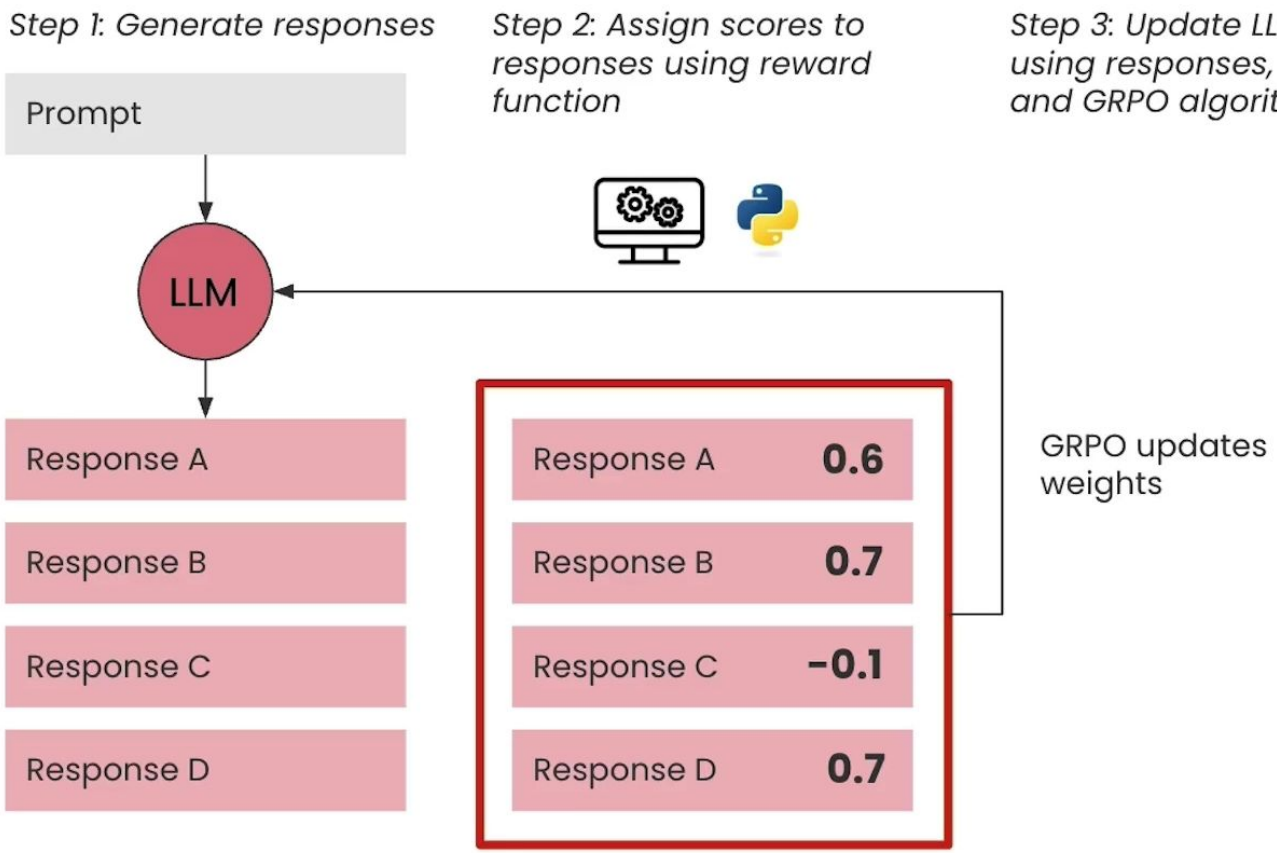
DPO: Contrastive Learning from Positive and Negative Samples



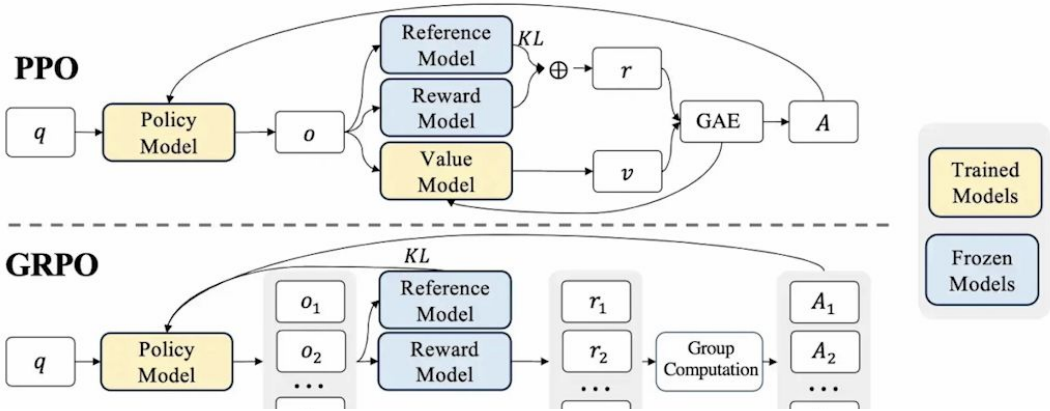
Challenges With RLHF and DPO

Challenge	RLHF	DPO
Data Needed	Ranked generations for reward model	Paired-preference labels (A > B)
Compute / Memory	Very high	Moderate
Training Stability	Often unstable—reward hacking, collapse risk	More stable, but still needs lots of labels
Limitation	Doesn't teach new tasks—only steers preferences	Doesn't teach new tasks—only steers preferences

Group Relative Policy Optimization (GRPO)



Policy Training in Online RL



$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{q,\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left(r_t(\theta)\hat{A}_{i,t}, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_{i,t}\right)\right] - \beta D_{\text{KL}}(\pi_\theta||\pi_{\text{ref}}).$$

The Objective (Equation)

- "Formally, the GRPO objective is:

$$J_{GRPO}(\theta) = \mathbb{E}_{q,\{o_i\}^G \sim \pi_{\theta_{old}}} \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_t(\theta) \hat{A}_{i,t},\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_{i,t} \right) \right] - \beta D_{KL}(\pi_\theta \,\|\, \pi_{ref})$$

- The first big term: average over the group of responses, and over the tokens in each response.
- The ratio $r_t(\theta)$ compares new Policy vs. old Policy probabilities, same as PPO.
- The **clip** ensures stable updates.
- The KL penalty keeps us close to the Reference."

GRPO is well suitable for Verifiable Rewards

Math: Check if the response matches ground truth

Prompt: What is 1+1-1+1.1-1
Response: The answer is \box{1.1}. ✅
Ground truth: 1.1

Coding: Running unit tests

Prompt: Given a string S, return the longest substring that occurs at least twice.
Response: import …
Test Input 1: “ABCDABCDBC”
Test Output 1: “ABCD”

- Requires preparation of ground truth for math, unit tests for coding, or sandbox execution environment for multi-turn agentic behavior
- More reliable than reward model in those domains
- Used more often for training reasoning models

GRPO vs PPO

- Both GRPO and PPO are very effective online RL algorithms!

- **GRPO:**
  - Well-suited for binary (often correctness-based) reward
  - Requires larger amount of samples
  - Requires less GPU memory (no value model needed)

- **PPO:**
  - Works well with reward model or binary reward
  - More sample efficient with a well-trained value model
  - Requires more GPU memory (value model)